Assignment 1

Q1. Introduction to the operating system with key functions.

An Operating System (OS) is system software that acts as an interface between users and computer hardware.
It manages hardware resources such as CPU, memory, storage, and input/output devices.
The OS also provides a platform for running applications efficiently and securely.
Key functions include process management, memory management, file management, and device control.

Q2) Introduction to the Unix/Linux (Architecture)

Unix/Linux architecture follows a layered design that organizes system functionality.
At the core is the kernel, which manages CPU, memory, devices, and system calls.
Above it is the shell, acting as the user interface to interact with the system.
On top are utilities and applications that perform user-level tasks.

Q3) Concept of Shell

The **shell** is a command-line interface that allows users to interact with the Unix/Linux operating system.
It interprets user commands and passes them to the kernel for execution.
The shell also supports scripting, enabling automation of tasks.
Common shells include **sh**, **bash**, **csh**, and **zsh**.

Q4) Types of Shell.

There are two main types of shells in Unix/Linux: Command-Line Shell (CLI) and Graphical Shell (GUI).
CLI shells include sh, bash, csh, ksh, and zsh, which use text-based commands.
GUI shells provide graphical interaction using windows, icons, and menus.
CLI shells are preferred for scripting and system administration tasks.

q5) Command structure.
A command structure in Unix/Linux defines how a command is written and executed.
It generally follows the format: command [options] [arguments].
The command specifies the action, options modify its behavior, and arguments define

the target.
Example: `ls -l /home` lists files in long format in the specified directory.

Q6) Introduction of basic linux commands (sudo, ls, pwd, mkdir, rmdir, rm, cd, cp, wc, mv, cmp,
passwd, who, uname)

Sudo - sudo is super user do , which allows admin access and it always asks for password to allow access

```
~ sudo pacman -Syu
[sudo] password for kakarot:
:: Synchronizing package databases...
 core is up to date
 extra is up to date
 multilib is up to date
:: Starting full system upgrade...
warning: btrfsmaintenance: local (1:0.5.2-1) is newer than extra (0.5.2-3)
 there is nothing to do
```

ls -is a Unix/Linux command used to list files and directories in the current or specified directory.

```
~/Doc/thapar_cs_lab  master ?1  ls
ds_lab  os  README.md
~/Doc/thapar_cs_lab  master ?1
```

pwd: A command that prints the full path of the current working directory in a terminal.

```
~ pwd
/home/kakarot
```

mkdir: A command used to create a new directory (folder).
rmdir: A command used to remove an empty directory.

```
~/Doc/thapar_cs_lab  master  ls
ds_lab  README.md
~/Doc/thapar_cs_lab  master  mkdir os
~/Doc/thapar_cs_lab  master  ls
ds_lab  os  README.md
~/Doc/thapar_cs_lab  master  rmdir os
~/Doc/thapar_cs_lab  master  ls
ds_lab  README.md
```

rm: A command used to delete files or directories.

```
~/Doc/thapar_cs_lab/os  master ?1  ls
file.txt
~/Doc/thapar_cs_lab/os  master ?1  rm file.txt
~/Doc/thapar_cs_lab/os  master  ls
~/Doc/thapar_cs_lab/os  master
```

cd: A command used to change the current directory.

```
~/Doc/thapar_cs_lab  master  cd os
~/Doc/thapar_cs_lab/os  master
```

cp: A command used to copy files or directories.

```
~/Doc/thapar_cs_lab  master ?1  ls
ds_lab  file1.txt  os  README.md
~/Doc/thapar_cs_lab  master ?1  cp file1.txt -t os
~/Doc/thapar_cs_lab  master ?2  ls
ds_lab  file1.txt  os  README.md
~/Doc/thapar_cs_lab  master ?2  cd os
~/Doc/thapar_cs_lab/os  master ?2  ls
file1.txt
~/Doc/thapar_cs_lab/os  master ?2
```

wc:wc is a Unix/Linux command used to count lines, words, and characters in a file or input.

```
~/Doc/thapar_cs_lab/os  master ?2  wc file1.txt
  21  223 1381 file1.txt
~/Doc/thapar_cs_lab/os  master ?2
```

mv- mv is a Unix/Linux command used to move or rename files and directories.

```
~/Doc/thapar_cs_lab/os  master ?2  mv file1.txt -t ../
~/Doc/thapar_cs_lab/os  master ?1  ls
~/Doc/thapar_cs_lab/os  master ?1  cd ../
~/Doc/thapar_cs_lab  master ?1  ls
ds_lab  file1.txt  os  README.md
~/Doc/thapar_cs_lab  master ?1
```

cmp- cmp is a Unix/Linux command used to compare two files and report the first difference between them.

```
~/Doc/thapar_cs_lab/os  master ?1  cmp file1.txt file2.txt
file1.txt file2.txt differ: byte 1, line 1
~/Doc/thapar_cs_lab/os  master ?1
```

Passwd- passwd is a Unix/Linux command used to change a user's password.

```
~/Doc/thapar_cs_lab/os  master ?1  passwd
Changing password for kakarot.
Current password:
passwd: Authentication failure
passwd: password unchanged
```

Who- who is a Unix/Linux command used to display users currently logged into the system.

uname -uname is a Unix/Linux command used to display system information such as the operating system and kernel.

```
~/Doc/thapar_cs_lab/os  master ?1  who
kakarot  seat0        2026-01-12 13:47 (login screen)
kakarot  tty2         2026-01-12 13:47 (tty2)
~/Doc/thapar_cs_lab/os  master ?1  whoami
kakarot
~/Doc/thapar_cs_lab/os  master ?1  uname
Linux
```

q7. How to install, update, upgrade and remove any package in linux (apt-get)

In Linux, apt-get is used to manage software packages from repositories.
Install a package using apt-get install package_name and remove it using apt-get remove package_name.
Update the package list with apt-get update and upgrade installed packages using apt-get upgrade.
These commands require administrator (sudo) privileges to execute.

```
~  sudo pacman -Syu
[sudo] password for kakarot:
:: Synchronizing package databases...
 core is up to date
 extra is up to date
 multilib is up to date
:: Starting full system upgrade...
warning: btrfsmaintenance: local (1:0.5.2-1) is newer than extra (0.5.2-3)
 there is nothing to do
```

Here we are using pacman because thats package manager for arch based systems

q8) > , >> option for directing the output of a command

> is used to redirect command output to a file by overwriting existing content.
>> is used to redirect output by appending it to the end of a file.
Both operators help store command results instead of displaying them on the screen.

q9) cat command.

The cat command (short for *concatenate*) is a basic Unix/Linux utility used to display the contents of files in the terminal.
It can combine multiple files and output them as a single stream.
Cat is often used for quick file viewing, file creation, and redirecting content to another file.
It works well with pipes to pass file contents to other commands.

q10) Compressing and archiving files (zip, tar)

Compressing and archiving files is used to reduce file size and bundle multiple files into one.
The tar command archives files together, and when combined with tools like **gzip** or **bzip2**, it also compresses them.

The zip command both compresses and archives files in a single step.
These tools make storage, transfer, and backups more efficient.

```
~/Documents/thapar_cs_lab/os  master ?2  ls
file  file1.txt  file2.txt
~/Documents/thapar_cs_lab/os  master ?2  zip zippy.txt file1.txt
  adding: file1.txt (deflated 42%)
~/Documents/thapar_cs_lab/os  master ?2  ls
file  file1.txt  file2.txt  zippy.txt
~/Documents/thapar_cs_lab/os  master ?2  tar -cvf archive.tar file1.txt
file1.txt
~/Documents/thapar_cs_lab/os  master ?2  ls
archive.tar  file  file1.txt  file2.txt  zippy.txt
~/Documents/thapar_cs_lab/os  master ?2
```